

Megan Ying

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Mechanical Engineering Experience

- Apple Interconnect Product Design Co-op** *Cupertino, CA | Sep 2023 – Aug 2024*
- Designed and integrated 4 connector systems for next-generation consumer devices in Siemens NX, meeting mechanical performance and manufacturing constraints while reducing projected unit costs by up to 22%
 - Drove connector production readiness by preparing 2D drawings and BOMs, reviewing Cpk results and first article inspections, and leading supplier and cross-functional DFM reviews
 - Performed tolerance stackup analysis and FEA to validate connector fit across manufacturing tolerances and structural performance under worst-case loads
 - Led connector troubleshooting and failure resolution, using DOE and root-cause analysis to guide design improvements and establish validation procedures
 - Automated analysis of 1,000+ Instron datasets using Python to identify design parameters that minimized connector clearance while meeting insertion-force requirements
- Caterpillar Hydraulics and Controls Engineering Intern** *Clayton, NC | May 2022 – Aug 2022*
- Characterized hydraulic lift, lower, and tilt performance across small-wheel-loader operating modes and recommended control changes to reduce cycle times
 - Developed Python and Power BI diagnostic tools to automate fault detection, identifying 8 high-priority machines for engineering investigation

Projects

- Battery Door** *Nov 2024 – Dec 2024*
- Designed a 9-component, one-touch battery-door release and re-latching mechanism in Siemens NX, using tolerance analysis, hand calculations, and FEA to validate worst-case engagement and 1,000-cycle durability
- Trash Compactor** *Sep 2024 – Dec 2024*
- Designed and fabricated an automated trash compactor in SolidWorks, integrating a machined scissor linkage, 3D-printed mounts, and Arduino controls into a functional electromechanical prototype
- Tripod Assistive Technology** *Nov 2022 – Dec 2022*
- Designed and prototyped an assistive tripod attachment in SolidWorks that automatically secures a phone using IR sensing and servo actuation, improving accessibility for users with limited hand strength

Additional Experience

- Bloomberg Software Engineer** *New York, NY | Oct 2025 – Present*
- Redesigned deployment processes for data pipelines supporting 250K+ securities, reducing manual steps by 40% while lowering deployment delays and risk of human error
 - Identified and resolved a distributed-database capacity risk through failure and usage analysis, redesigning the data architecture and verifying data integrity across 28.3M migrated records
- Design Research Collective Research Assistant** *Pittsburgh, PA | Sep 2024 – Jul 2026*
- Published 2 first-author, peer-reviewed papers examining how reasoning strategy and model size affect AI agreement with expert engineering material-selection decisions
 - Developed and open-sourced a simulated-annealing module for constrained engineering design optimization
- UBS Quantitative Trading Summer Analyst** *New York, NY | Jun 2023 – Aug 2023*
- Developed Python tools to evaluate pricing strategies and monitor risk across 1,000+ clients, enabling earlier loss detection and data-driven model selection

Education

- Carnegie Mellon University (CMU)** *Pittsburgh, PA | May 2025*
B.S. in Mechanical Engineering — GPA: 4.0

Skills

CAD & Design: Siemens NX, SolidWorks, GD&T, DFM | **Analysis:** Tolerance Analysis, FEA, ANSYS, DOE, Failure Analysis
Manufacturing: Injection Molding, Manual Machining, Stamping, 3D Printing | **Programming:** Python, Arduino, MATLAB, C